A continuous function f is weakly-analytic on a region $\Omega \subset \mathbb{C}$ if

$$\int_\Omega f \frac{\partial \phi}{\partial \bar{z}} \, \mathrm{d}A = 0,$$

for $\phi \in C_c^1(\Omega)$. Here dA denotes the area element. Prove that a continuous function is weakly-analytic iff it is analytic.

Proof

(i) If f is analytic, then by definition $\frac{\partial f}{\partial \bar{z}} = 0$. Hence

$$\frac{\partial}{\partial \bar{z}}(f\phi) = \frac{\partial f}{\partial \bar{z}}\phi + \frac{\partial \phi}{\partial \bar{z}} f = \frac{\partial \phi}{\partial \bar{z}} f.$$

By the complex version of Green's theorem (since $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$),

$$\int_\Omega f \frac{\partial \phi}{\partial \bar{z}} \, \mathrm{d}A = \int_\Omega \frac{\partial}{\partial \bar{z}}(f\phi) \, \mathrm{d}A = \frac{1}{2i} \int_{\partial\Omega} f\phi \, \mathrm{d}z = 0$$

since ϕ has compact support, $f\phi = 0$ on $\partial\Omega$.
(ii) Conversely, suppose f is weakly-analytic. We construct a sequence f_ε such that f_ε are holomorphic and $f_\varepsilon \to f$ uniformly as $\varepsilon \to 0$, then clearly f is holomorphic.
Consider a non-negative test function $\phi \in C_c^\infty(\mathbb{R}^2)$ with $\mathrm{supp}\phi \subset \mathbb{D}$ and $\phi(-z) = \phi(z)$, such that $\int_{\mathbb{C}} \phi(z) \, \mathrm{d}A = 1$, and define $\phi_\varepsilon(z) = \frac{1}{\varepsilon^2}\phi\left(\frac{z}{\varepsilon}\right)$, then clearly $\int_{\mathbb{C}} \phi_\varepsilon \, \mathrm{d}A = 1$. Define

$$f_\varepsilon(z) = (f * \phi_\varepsilon)(z) = \int_\Omega f(w)\phi_\varepsilon(z-w) \, \mathrm{d}A.$$

Then it is well known that $f_\varepsilon \in C_c^\infty(\Omega)$. Note that

$$\frac{\partial}{\partial \bar{z}} f_\varepsilon(z) = \int_\Omega \frac{\partial}{\partial \bar{z}}(f(w)\phi_\varepsilon(z-w)) \, \mathrm{d}A = -\int_\Omega f(w) \frac{\partial}{\partial \bar{w}} \phi_\varepsilon(z-w) \, \mathrm{d}A.$$

For sufficiently small ε, $\frac{\partial}{\partial \bar{w}}\phi_\varepsilon(z-w)$ is a C^1 function with compact support in Ω. Since f is weakly-analytic, $\frac{\partial}{\partial \bar{z}} f_\varepsilon = 0$ hence f_ε is holomorphic. Therefore the uniform limit f of f_ε is also holomorphic.

Connection with Weyl's lemma

The modern version of Weyl's lemma is:
Theorem: Let $\Omega \subset \mathbb{R}^n$, a function $u \in L^1_{loc}(\Omega)$ is weakly harmonic if

$$\int_\Omega u \, \Delta\varphi \, dx = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

Then u is weakly harmonic iff u is (smoothly) harmonic.
Proof: (i) If u is harmonic, then

$$\int_\Omega u\Delta\varphi\,\mathrm{d}x = -\int_\Omega \nabla u\cdot\nabla\varphi\,\mathrm{d}x = \int_\Omega \varphi\Delta u\,\mathrm{d}x = 0.$$

(ii) Conversely suppose u is weakly harmonic, we again consider the standard symmetric mollifier $\phi \in C_c^\infty(\mathbb{R}^n)$ such that $\phi \geqslant 0$, $\mathrm{supp}\phi \subset B(0,1), \phi(x) = \phi(-x)$, and $\|\phi\|_{L^1} = 1$. Let $\phi_\varepsilon(x) = \frac{1}{\varepsilon^n}\phi\left(\frac{x}{\varepsilon}\right)$ and $\Omega_\varepsilon = \{x \in \Omega : d(x, \overline{\Omega}) > \varepsilon\}$. Then $u_\varepsilon = u * \phi_\varepsilon$ is smooth on Ω_ε, so

$$\Delta u_\varepsilon(x) = \int_\Omega u(y)\Delta_x\phi_\varepsilon(x-y)\,\mathrm{d}y = -\int_\Omega u(y)\Delta_y\phi_\varepsilon(x-y)\,\mathrm{d}y = 0, \quad \text{for all } x \in \Omega_\varepsilon.$$

since $\Delta\phi_\varepsilon$ is a $C_c^\infty(\Omega)$ function. Therefore u_ε is harmonic in Ω_ε, and it is well known that since $u \in L^1_{loc}(\Omega)$, we have $u_\varepsilon \to u$ in $L^1_{loc}(\Omega)$. Recall u is harmonic iff u satisfy the mean value property, so L^1_{loc} convergence implies the Cauchy condition for locally uniform convergence. Therefore u is the uniform limit of u_ε, and u is harmonic.

More generally, the same result holds for distributional solutions of Laplace's equation.